Year 8 Mathematics

Unit 1: Number

1.2 Prime Factor Decomposition

Prime Factor Decomposition

STARTER · 5 MIN

Date: _____ **Title: Prime Factor Decomposition**

LEARNING OBJECTIVES

- To be able to write the prime factor decomposition of a number
- To be able to use prime factor decomposition to find the HCF and LCM

STARTER *Answer in your book. No calculators.*

1. Write the factors of 16
2. Write the factors of 28
3. Write the first 5 prime numbers
4. Is 1 a prime number?

Starter — Answers

STARTER

Tick or correct in green pen

1. **1, 2, 4, 8, 16**
2. **1, 2, 4, 7, 14, 28**
3. **2, 3, 5, 7, 11**
4. **No: it has one factor**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can break down a composite number into its prime factors using a factor tree	☐
SILVER	I can express prime factorisation in index form	☐
GOLD	I can find the HCF and LCM of two numbers and use them to solve problems	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Prime Factor Decomposition

INTRODUCTION · 3 MIN

Every whole number above 1 can be written as a product of prime numbers.

Prime number

Exactly two factors: 1 and itself.

2, 3, 5, 7, 11, 13, ...

Prime factor decomposition

A product of primes, in index form.

$60 = 2^2 \times 3 \times 5$

HCF

Multiply the primes both numbers share.

HCF of 12 and 18 = 2 × 3 = 6

LCM

Multiply every prime, shared ones once.

LCM of 12 and 18 = 36

Examples 1 and 2 — Factor Trees

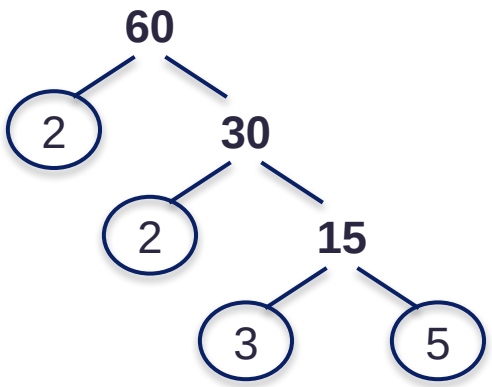
EXAMPLES · 5 MIN

BRONZE *Watch and copy into your book*

Hint Split until every branch is prime Primes in order

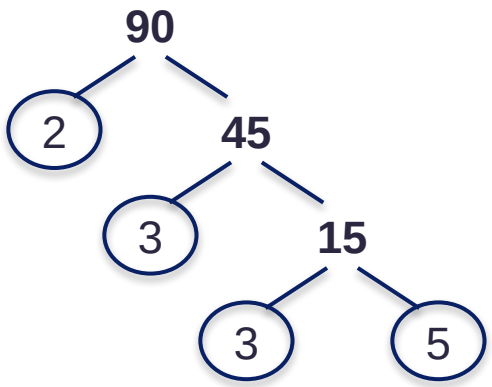
Example 1

Write 60 as a product of its prime factors.



Example 2

Write 90 as a product of its prime factors.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Split until every branch ends in a prime

$60 = 2 \times 2 \times 3 \times 5$

$= \mathbf{2^2 \times 3 \times 5}$

Tip: write the primes in order.

Example 2

Split until every branch ends in a prime

$90 = 2 \times 3 \times 3 \times 5$

$= \mathbf{2 \times 3^2 \times 5}$

Tip: check: $2 \times 9 \times 5 = 90$.

Task 1

PRACTICE · 9 MIN

BRONZE

Write as a product of prime factors

Hint Split until every branch is prime Primes in order

1. 24
2. 36
3. 75
4. 54
5. 32
- ★ 140

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $2^3 \times 3$

2. $2^2 \times 3^2$

3. 3×5^2

4. 2×3^3

5. 2^5

★ $2^2 \times 5 \times 7$

Examples 3 and 4 — Index Form and Bigger Numbers

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Repeated prime: use a power $2 \times 2 \times 2 = 2^3$

Example 3

Write 450 as a product of prime factors in index form.

Example 4

$630 = 2 \times 3^2 \times 5 \times 7$

Use this to write 1 260 as a product of prime factors.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Divide by primes, smallest first

$450 = 2 \times 225$

$= 2 \times 3 \times 3 \times 5 \times 5$

$\mathbf{= 2 \times 3^2 \times 5^2}$

Tip: 225 = 9 × 25.

Example 4

1 260 = 2 × 630

$1\,260 = 2 \times 2 \times 3^2 \times 5 \times 7$

$\mathbf{= 2^2 \times 3^2 \times 5 \times 7}$

Tip: one more 2, so 2 becomes 2^2.

Task 2

PRACTICE · 9 MIN

SILVER *Write in index form*

Hint Repeated prime: use a power $2 \times 2 \times 2 = 2^3$

1. 225
2. 420
3. 92
4. 160
5. 156

★ 441

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $3^2 \times 5^2$

2 $2^2 \times 3 \times 5 \times 7$

3 $2^2 \times 23$

4 $2^5 \times 5$

5 $2^2 \times 3 \times 13$

★ $3^2 \times 7^2$

Examples 5 and 6 — HCF and LCM from Prime Factors

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint HCF: shared primes LCM: all primes once

Example 5

(3 marks)

$90 = 2 \times 3^2 \times 5$

$252 = 2^2 \times 3^2 \times 7$

Find the HCF and the LCM of 90 and 252.

Example 6

(2 marks)

Buses leave every 12 min, trams every 18 min. Both leave at 8 am. When do they next leave together?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

HCF: shared primes. LCM: all primes

HCF $= 2 \times 3^2$

$= 18$

LCM $= 2^2 \times 3^2 \times 5 \times 7$

$= 1\,260$

Tip: use the highest power of each prime.

Example 6

'Together again' means LCM

$12 = 2^2 \times 3$, $18 = 2 \times 3^2$

LCM $= 2^2 \times 3^2 = 36$

Time = 8:36 am

Tip: greatest number of groups means HCF.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: HCF and LCM

Hint HCF: shared primes LCM: all primes once

1. HCF and LCM of 12 and 18

2. HCF and LCM of 12 and 20

3. HCF and LCM of 15 and 40

4. HCF and LCM of 24 and 36

5. 30 pens and 42 pencils shared into equal packs. Greatest number of packs?

★ Lights flash every 6 s and 8 s. When do they next flash together?

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. **HCF 6, LCM 36**
2. **HCF 4, LCM 60**
3. **HCF 5, LCM 120**
4. **HCF 12, LCM 72**
5. **6 packs (HCF)**

★ **After 24 s (LCM)**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE True or false: 60 = 2 × 3 × 10

SILVER Write 48 as a product of prime factors

GOLD Find the HCF and LCM of 12 and 20

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE

False: 10 is not prime. $60 = 2^2 \times 3 \times 5$

SILVER

$2^4 \times 3$

GOLD

HCF = 4, LCM = 60

Exam Practice — Prime Factors

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint Split until every branch is prime
Primes in order

Question 1 (2 marks)

Write 36 as a product of its prime factors.

.....

Question 2 (2 marks)

Write 84 as a product of its prime factors.

.....

Exam Practice — Index Form

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Repeated prime: use a power $2 \times 2 \times 2 = 2^3$

Question 3

(2 marks)

Write 360 as a product of prime factors.
Give your answer in index form.

.....

Question 4

(2 marks)

$140 = 2^2 \times 5 \times 7$

Write 420 as a product of prime factors in index form.

.....

Exam Practice — HCF and LCM

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint HCF: shared primes LCM: all primes once

Question 5

(2 marks)

$72 = 2^3 \times 3^2$ and $120 = 2^3 \times 3 \times 5$

(a) Find the HCF of 72 and 120. (1)

.....

(b) Find the LCM of 72 and 120. (1)

.....

Question 6

(3 marks)

Ali visits the gym every 6 days.
Sara visits it every 10 days.
They both go on 1 March.

On what date will they next both go?

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$2 \times 2 \times 3 \times 3$ — M1

$2^2 \times 3^2$ — A1

Tip: a factor tree helps.

Question 2

$2 \times 2 \times 3 \times 7$ — M1

$2^2 \times 3 \times 7$ — A1

Tip: 84 = 4 × 21.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$2 \times 2 \times 2 \times 3 \times 3 \times 5$ — M1

$2^3 \times 3^2 \times 5$ — A1

Tip: 360 = 8 × 45.

Question 4

$420 = 3 \times 140$ — M1

$2^2 \times 3 \times 5 \times 7$ — A1

Tip: put the primes in order.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) $2^3 \times 3 = 24$ — B1

(b) $2^3 \times 3^2 \times 5 = 360$ — B1

Tip: HCF shared, LCM everything.

Question 6

$6 = 2 \times 3,\ 10 = 2 \times 5$ — M1

LCM = 30 days — M1

31 March — A1

Tip: 1 March + 30 days.